S. Andrew Lanham

Contact: andrewlanham55@gmail.com Portfolio: https://andrewlanham.github.io/

SUMMARY

I enjoy building, benchmarking and improving performance of signal processing systems in an explainable manner. I have nine years of professional experience in quantum computing research, signal processing, probabilistic modeling, and hardware/firmware bring-up – most recently as engineering lead on an automated commercial hardware product now in the field. My work covers the research process from literature review through proof of concept to deployment, and has produced fifteen publications in quantum information, wireless communication, and biomedical modeling, and one patent.

SKILLS

- **Software**: Python, MATLAB, C/C++, Git, simulation and numerical algorithm validation, AI-assisted tools (Claude Code), NumPy/SciPy/pandas
- **Embedded**: PIC32, STM32F4, bare-metal firmware, C/C++, fixed-point arithmetic, I2C, SPI, UART, real-time digital I/O, peripheral drivers, MPLAB X, Keil
- **Hardware**: PCB design, sensor design, mechanical prototyping, evaluation boards, supplier and manufacturing coordination
- **Lab & Test**: oscilloscope, logic analyzer, spectrum analyzer, signal and function generators, bench supplies, board bring-up and debug, unit testing, hardware-in-the-loop validation, test procedures
- **Signal Processing & Estimation**: detection strategies, noise modeling, FFT/spectral analysis, Gaussian statistics, Bayesian inference, information-theoretic experimental design (MRI data), wireless communication, digital filtering
- **Quantum**: quantum error correction, fusion-based quantum computing, quantum resource simulation, quantum-inspired algorithms, benchmarking against quantum devices
- **Specification and Documentation**: technical specifications, interface definition, requirements documentation, stakeholder validation, Microsoft Excel, Microsoft PowerPoint

EDUCATION

The University of Texas at Austin, Austin, TX
Master of Science, Electrical Engineering, August 2019

- Thesis: Quantum Information Processing Approaches in Classical Systems (Advisors: Robert W. Heath Jr., Brian La Cour)

Bachelor of Science, Electrical Engineering, May 2017

EXPERIENCE

Engineering Lead and Hardware Design Sky High Cold Brew
Austin, TX January 2025 - Present

Hardware and firmware design to replace a keg-based product line, which relied on nitrogen tanks and pressure-fed tubing, with a post-mix generation: concentrate brewed and mixed at the point of dispense in a smaller enclosure. Minimum viable building size dropped from 150+ units to 50+, and the smaller system also fits commercial sites. Addressable market roughly 10x larger, at under 50% of the original BOM cost. Performed extensive failure testing and first units are now in the field with designs passed to larger-scale manufacturers.

- **Hardware Engineering**: PCB and embedded design, prototyping and deployment; pump, valve and mixing chamber integration; mechanical prototyping
- **Customer and Deployment**: Worked directly with site operators on deployment requirements, and with manufacturers on production constraints
- **Supply Chain**: sourced and qualified suppliers and set up the manufacturing chain behind the new line

Engineering Scientist Center for Quantum Research
Applied Research Laboratories, UT Austin August 2019 - January 2025

Led research and development at the intersection of quantum information science, advanced signal

processing, and analog hardware systems. My work included architecting novel algorithms and benchmarking analog and digital computing systems against quantum devices.

- **Benchmarking and Characterization**: Benchmarking analog computing systems. Modeling noise, detection strategies. MATLAB, Python, and C/C++ software library design for signal processing and statistical benchmarking
- **Statistical Modeling**: Classical models of quantum-classical boundary, Gaussian statistics and post-selection. Simulation and analysis software development in Python
- **Embedded Implementation**: Embedded system and PCB design using the STM32F4 family
- **Wireless Communication and Signal Processing**: New algorithms for wireless communication in the area of multiantenna space-time block coding. Algorithm design and validation in MATLAB, Python
- **Hardware Collaboration**: Performed Bayesian optimal experimental design with applications to hyperpolarized MRI and magnetic nanoparticle hyperthermia treatment, with work appearing in journal of Applied Mathematical Modelling. Contributed solver algorithms to Snap-SAT, a compute-in-memory SAT solver chip (ISSCC 2023).
- **Clearance**: Secret (inactive; eligible for reinstatement through January 2027)

Engineer, Systems Architecture Tacen, Inc.
November 2017 - January 2019

Authored the formal technical documentation and system specifications for a high-performance, non-custodial digital asset exchange by communicating with project engineers. I was responsible for mathematically modeling and documenting the core logic to guide external engineering and stakeholder validation.

- **System Specification**: Authored end-to-end technical spec for a decentralized exchange backend and asset custody system
- **Interface Definition**: Interfaced with engineering teams to define and document modular interfaces for DEX backend components

Graduate Research Assistant Center for Quantum Research
Applied Research Laboratories, UT Austin May 2017 - August 2019

Pioneered the experimental application of quantum error correction (QEC) to classical analog computing systems

Firmware Engineer and Product Design Modular Music Systems
November 2016 - September 2017

Designed and implemented the complete embedded firmware for a multi-channel digital/analog modular synthesizer module using a PIC32 microcontroller. My responsibilities included architecting the multi-mode user interface, managing real-time digital I/O, and controlling the I2C DAC for the analog output stages.

- **Embedded Design**: Designed PIC32-based firmware controlling a modular synthesizer's multi-mode envelope generation

ROLES AND MENTORSHIP

- **Science Communication**: Recently created *Quantum Computing Review*, a YouTube channel reviewing quantum optics literature for a general audience; also explores AI-assisted research and graphic design
- **Technical Communication and Global Presentations**: Delivered talks and project presentations at international conferences and showcases
- **University Tutor and Research Mentor**: Tutored advanced technical subjects and mentored students on undergraduate theses and internships

PATENTS

"Space-time encoding in wireless communication systems." U.S. Patent No. 11,218,195. 4 Jan. 2022.

SELECTED PUBLICATIONS

Nandyala, Mahesh and **S.A. Lanham** and Jha, Prashant K. and Wu, Chengyue and Hazle, John D. and Yankeelov, Thomas E. and Stafford, R. Jason and El-Gendy, Ahmed A. and Fuentes, David,

"An Information-Theoretic Framework for Optimal Experimental Design in Magnetic Nanoparticle Hyperthermia." *Applied Mathematical Modelling.* 2026.

S.A. Lanham, and Brian R. La Cour. "Quantum Resource States from Post-Selected Classical Mixed States." *2024 IEEE International Conference on Quantum Computing and Engineering (QCE).* Vol. 1. IEEE 2024.

S. Xie, M. Yang, **S.A. Lanham**, M. Wang, Y. Wang, S. Oruganti, J.R. Kulkarni. "Snap-SAT: A One-Shot Energy-Performance-Aware All-Digital Compute-in-Memory Solver for Large-Scale Hard Boolean Satisfiability Problems." *2023 IEEE International Solid-State Circuits Conference (ISSCC).* IEEE 2023.

T.C. Cuvelier, **S.A. Lanham**, B.R. La Cour, and R.W. Heath Jr. "Quantum Codes for Classical Communication: A Space-Time Block Code from Quantum Error Correction." *IEEE Open Journal of the Communications Society.* October 2021.

S.A. Lanham, E. Bradley, and B.R. La Cour. "Generalized Noncoherent Space-Time Block Codes from Quantum Error Correction." *2022 IEEE Military Communications Conference (MILCOM).* IEEE 2022.

ADDITIONAL PUBLICATIONS

S.A. Lanham, and Brian R. La Cour. "Bell State Statistics with Post-Selection Loophole." *Quantum 2.0.* Optica Publishing Group, 2024.

S.A. Lanham, and B.R. La Cour. "A Quantum-Inspired Classical Solver for Boolean k-Satisfiability Problems." *2021 IEEE International Conference on Quantum Computing and Engineering (QCE).* IEEE 2021.

S.A. Lanham, and B.R. La Cour. "Detection-Based Measurements for Quantum Emulation Devices." *2020 IEEE International Conference on Quantum Computing and Engineering (QCE).* IEEE 2020.

S.A. Lanham, T.C. Cuvelier, C. Ostrove, B.R. La Cour, G. Ott and R.W. Heath Jr. "A Noncoherent Space-Time Code from Quantum Error Correction." *2019 53rd Annual Conference on Information Sciences and Systems (CISS)*, Baltimore, MD, March 2019, pp. 1-6.

B.R. La Cour, G.E. Ott, and **S.A. Lanham**. "Using Quantum Emulation for Advanced Computation." *2017 IEEE Custom Integrated Circuits Conference (CICC)*, Austin, TX, 2017.

(*archival*) Lanham, S. Andrew. "Quantum-Inspired Approximations to Constraint Satisfaction Problems." arXiv preprint arXiv:2212.04016 (2022).

C. Ostrove, B.R. La Cour, **S.A. Lanham**, and Granville Ott. "Improving Performance of an Analog Electronic Device Using Quantum Error Correction." *Journal of Physics Communications.* 2019.

B.R. La Cour, **S.A. Lanham** and C. Ostrove. "Parallel Quantum Computing Emulation." *2018 IEEE International Conference on Rebooting Computing (ICRC)*, McLean, VA, 2018, pp. 1-9.